

How to draw a histogram

To draw a histogram, begin by making a frequency table for the data. Next, using the class boundaries, find the width of each class of data. Then calculate frequency density for each by dividing frequency by class width.

upper class boundary for any group is lower boundary of next group

class boundaries for this data are 10, 16, 19, 26, and 30

find class width by subtracting lower class boundary from the upper class boundary, for example $16 - 10 = 6$

number of downloads per month

divide frequency by class width to find frequency density

Age (year)	Frequency (downloads in a month)
10–15	12
16–18	15
19–25	28
26–29	12
>30	0

Age	Class width	Frequency	Frequency density
10–15	6	12	2
16–18	3	15	5
19–25	7	28	4
26–29	4	12	3
>30	–	0	–

there is no data to enter for this group

The information needed to draw a histogram is the range of each class of data and frequency data. From this information, the class width and frequency density can be calculated.

To find class width, begin by finding the class boundaries of each group of data. These are the two numbers that all the values in a group fall in between—for example, for the 10–15 group they are 10 and 16. Next, find class width by subtracting the lower boundary from the upper for each group.

To find frequency density, divide the frequency by the class width of each group. Frequency density shows the frequency of each group in proportion to its class width.

